



EG1113 - C Programming for Engineering Lab

Report #1

Virtual Machines, Hello World, and Eliza

Date of the experiment: August 21, 2024

Due date of report: August 23, 2021

Name: Nicholas Galus

1. Core concepts learned

- Setup remote desktop to utilize the assigned Raspberry Pi 400
- Established network file sharing between the Raspberry Pi 400 and laptop utilizing Caja
- Created a hello world program in the C programming language

2. List of equipment and software used

- Raspberry Pi 400
- Notepad
- Lenovo Laptop
- Caja File Network Share

3. Relevant web links and other documents:

- Utilized [W3Schools](#) to create a for loop to use fewer lines of code to print out the 10 lines of “Hello, this is Nick Galus's first C program on a PI 400 at STMU.”

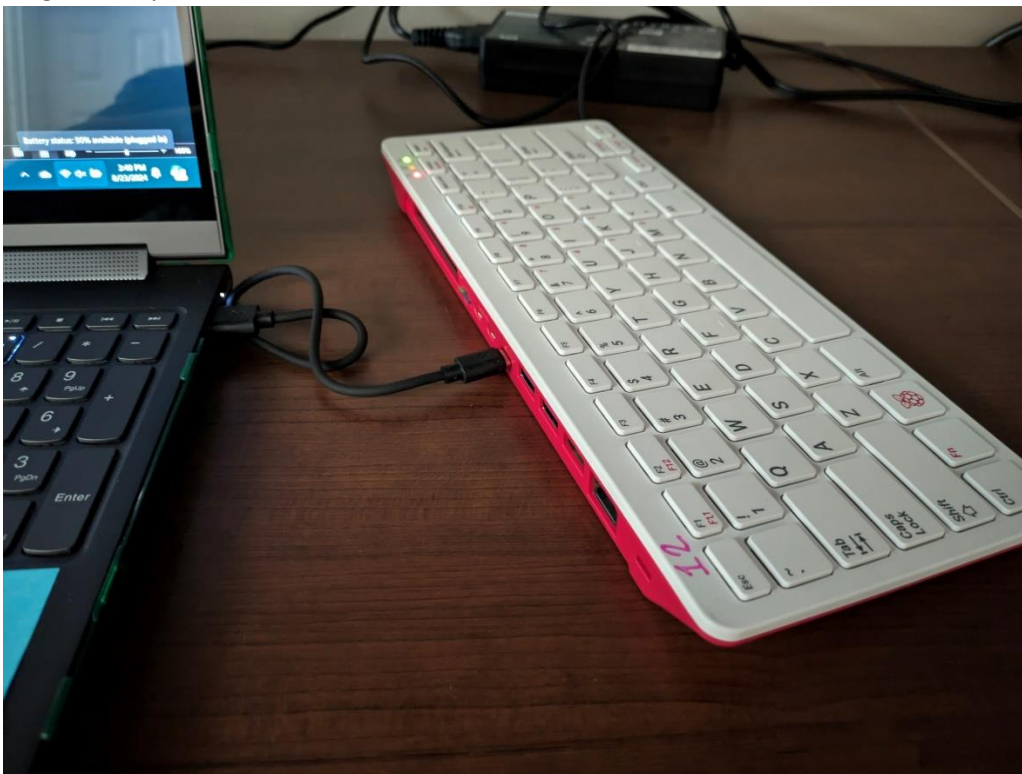
4. Exercise Raspberry Pi Set Up

4.1 Summary

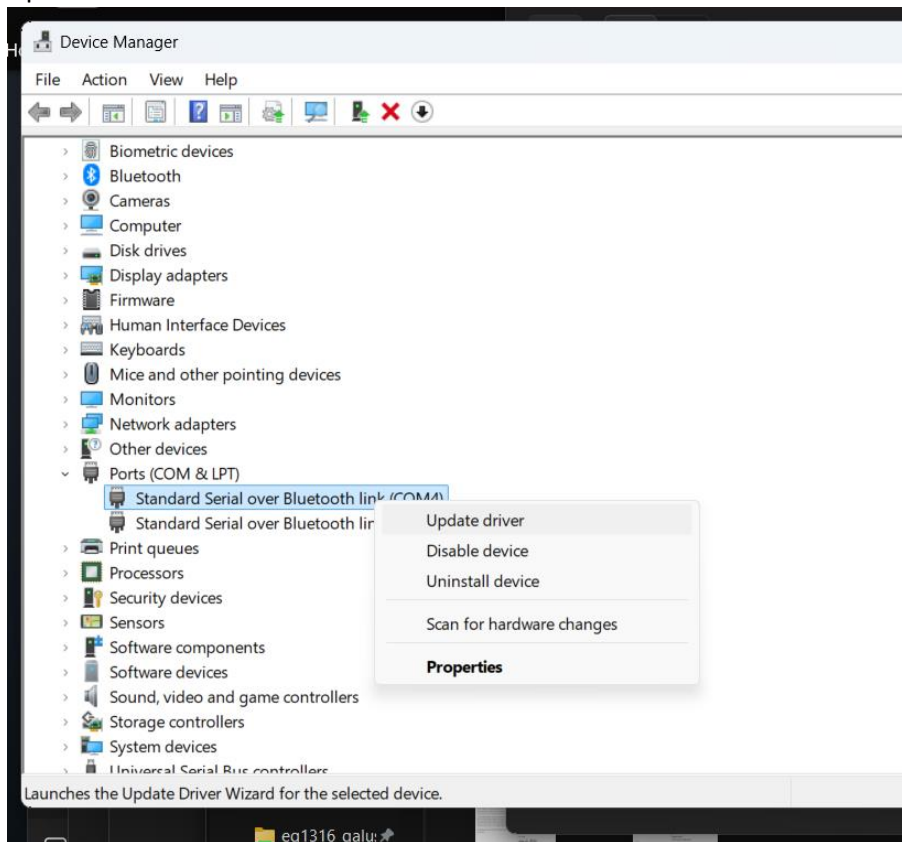
- Setup the Raspberry PI to be used with the Remote Desktop application

4.2 Procedures

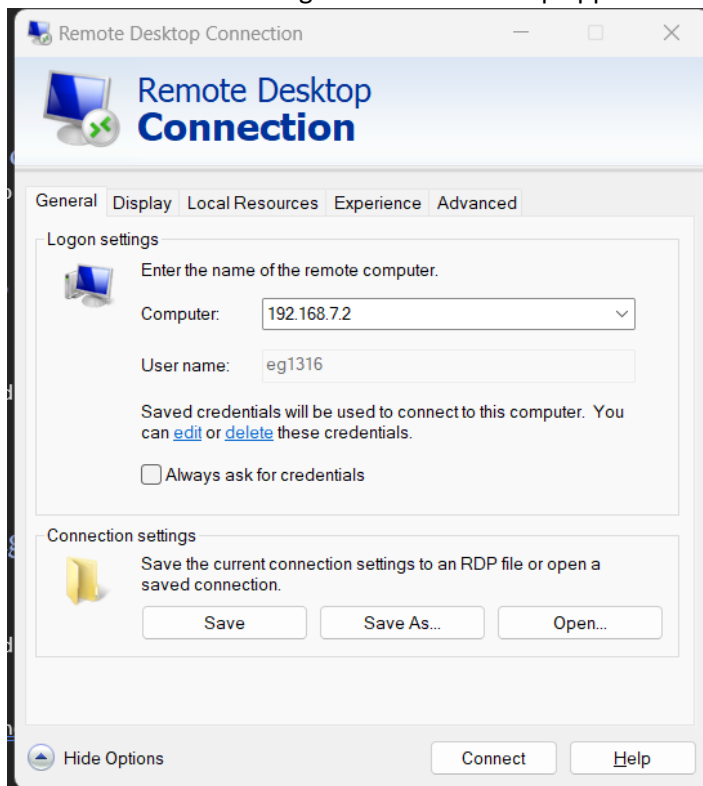
1. Plug PI in to personal device



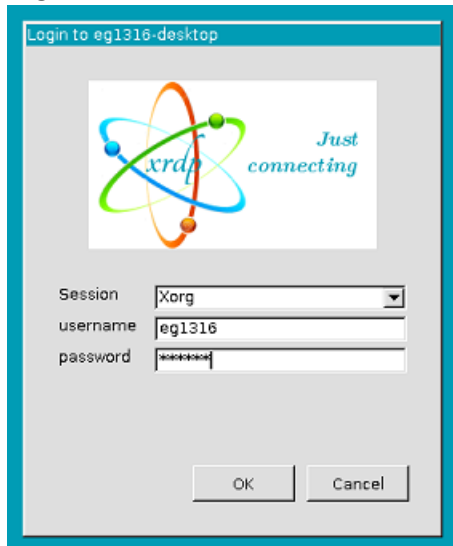
2. Update drivers for PI



3. Connect to the PI utilizing the remote desktop application



4. Login to the PI



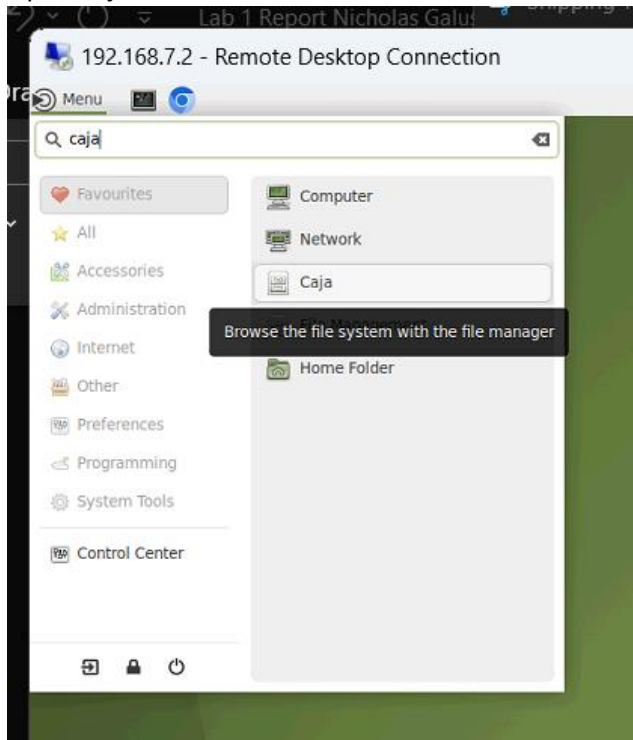
5. Exercise Network File Sharing

5.1 Summary

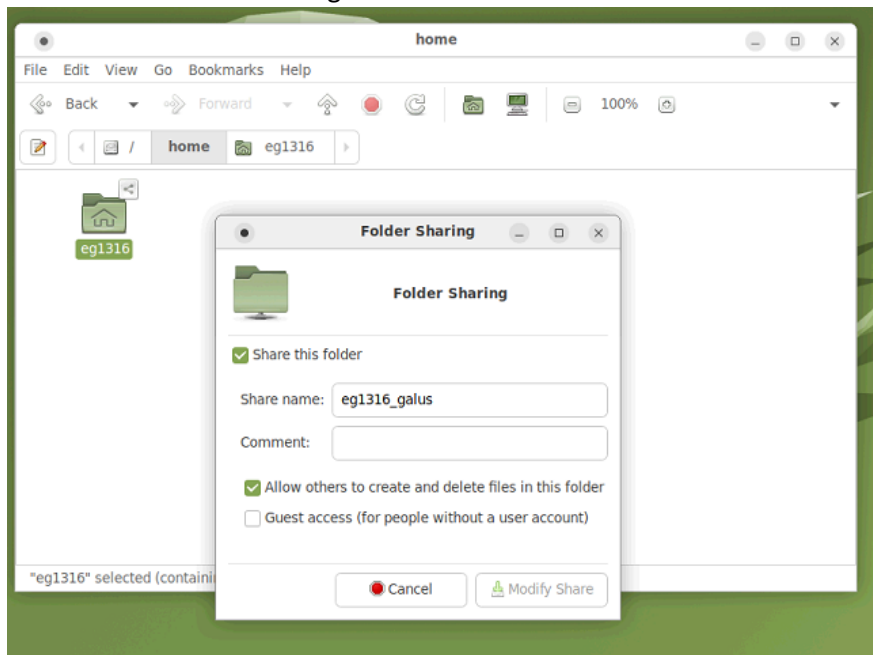
- Setup Network File Sharing between the personal device and PI utilizing Caja.
- An issue that occurred was the network file sharing protocol was disabled.
 - The resolution was to reenale it which prompted the sharing of files between the 2 devices.

5.2 Procedures

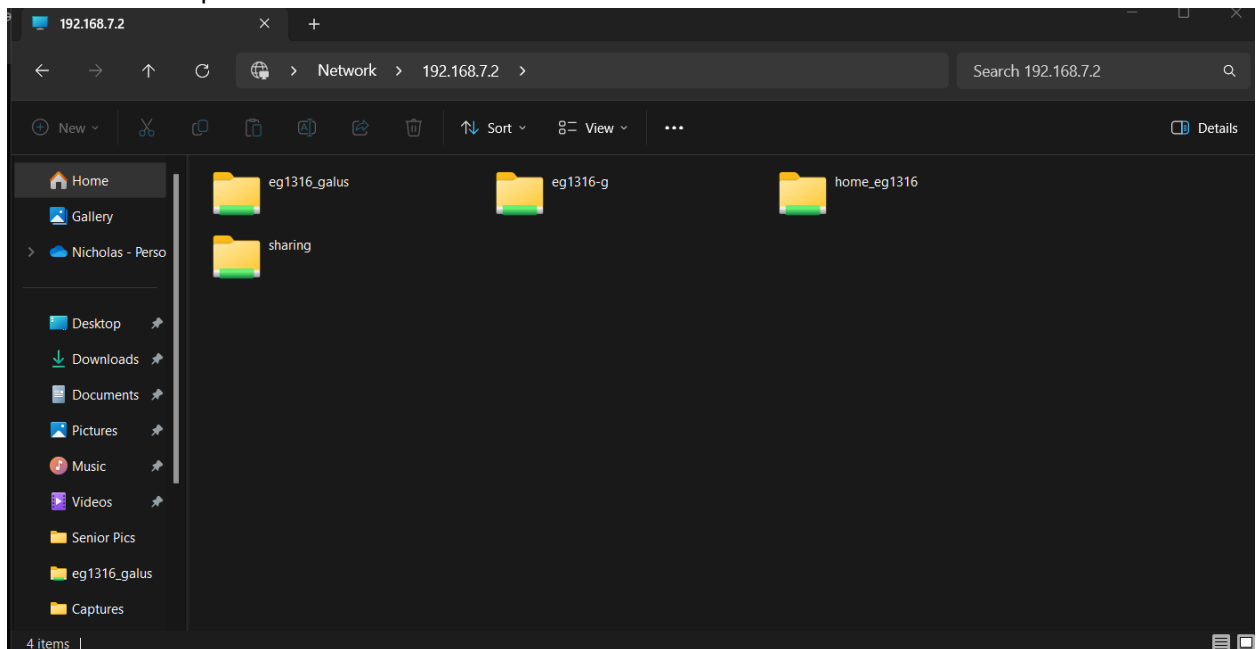
1. Open Caja on the PI



2. Enable Network File Sharing on the main drive of the PI



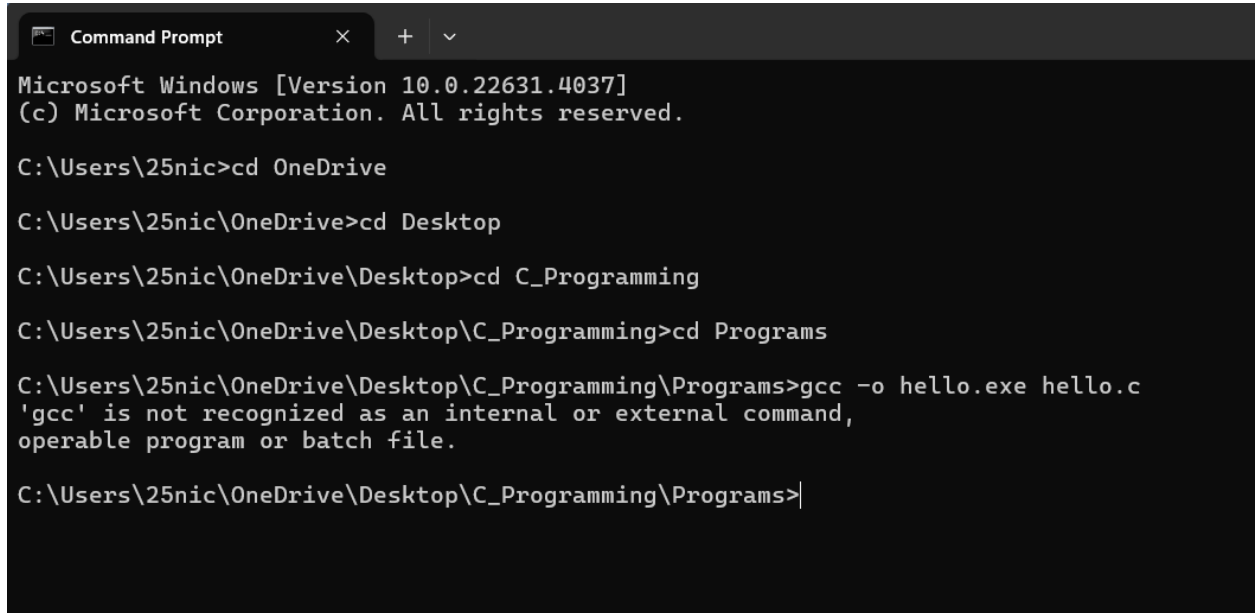
3. Access it on the personal device



6. Exercise Hello World Program

6.1 Summary

- Create a hello program in C to print “Hello, this is Nick Galus's first C program on a PI 400 at STMU.” Once executed successfully, modify it to print the same thing 10 times.
- An issue resulted when attempting to run the C program from the Windows command prompt.



```
Command Prompt
Microsoft Windows [Version 10.0.22631.4037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\25nic>cd OneDrive

C:\Users\25nic\OneDrive>cd Desktop

C:\Users\25nic\OneDrive\Desktop>cd C_Programming

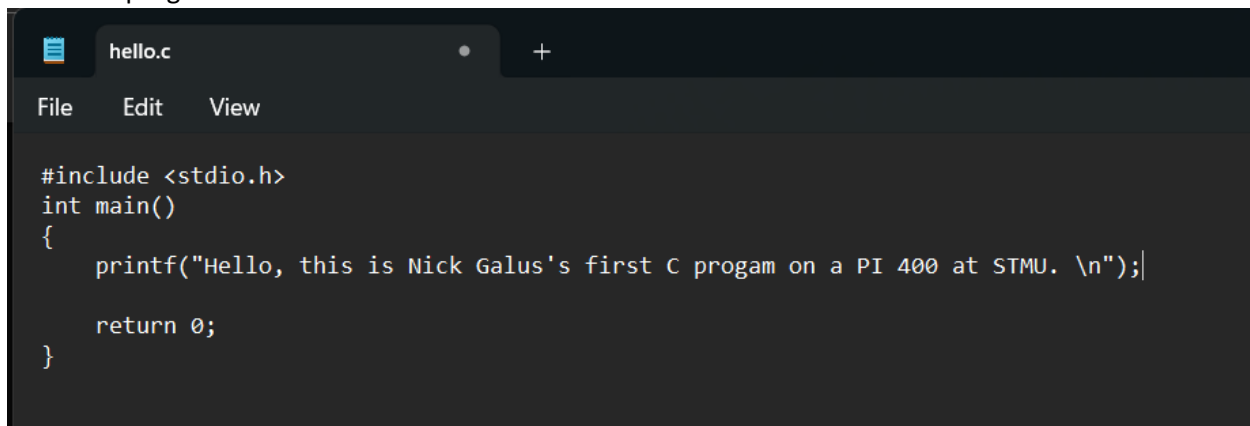
C:\Users\25nic\OneDrive\Desktop\C_Programming>cd Programs

C:\Users\25nic\OneDrive\Desktop\C_Programming\Programs>gcc -o hello.exe hello.c
'gcc' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\25nic\OneDrive\Desktop\C_Programming\Programs>|
```

6.2 Procedures

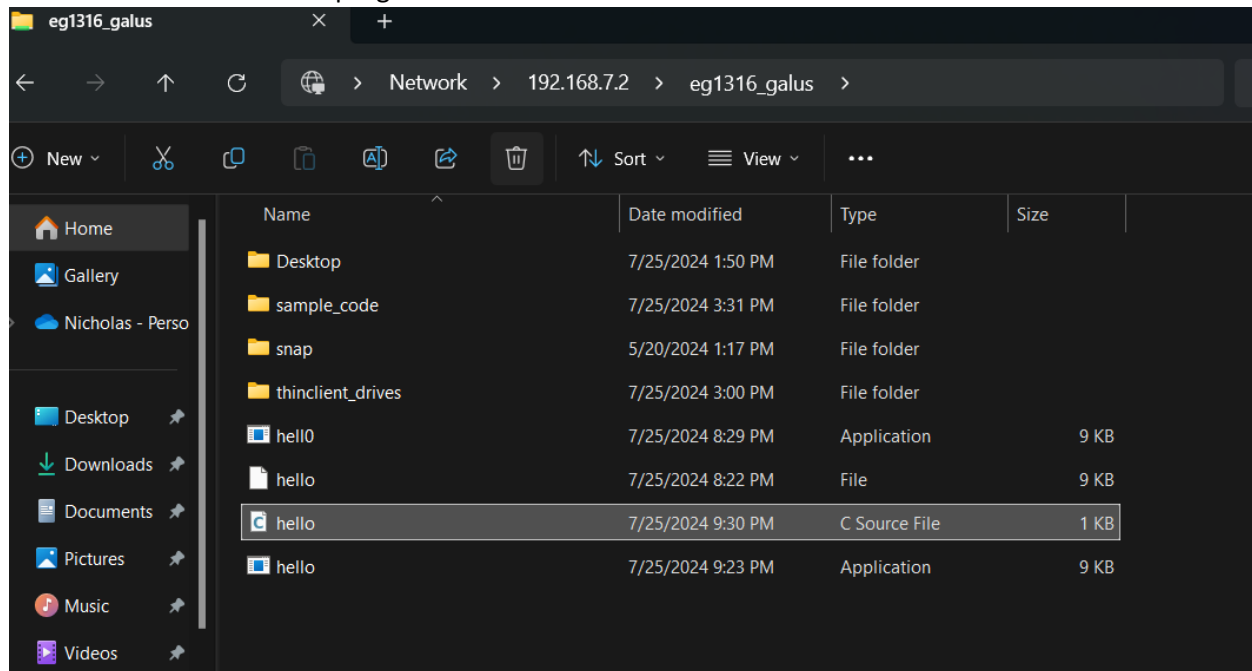
1. Create a “hello program” utilizing notepad named hello.c to print “Hello, this is Nick Galus's first C program on a PI 400 at STMU.”



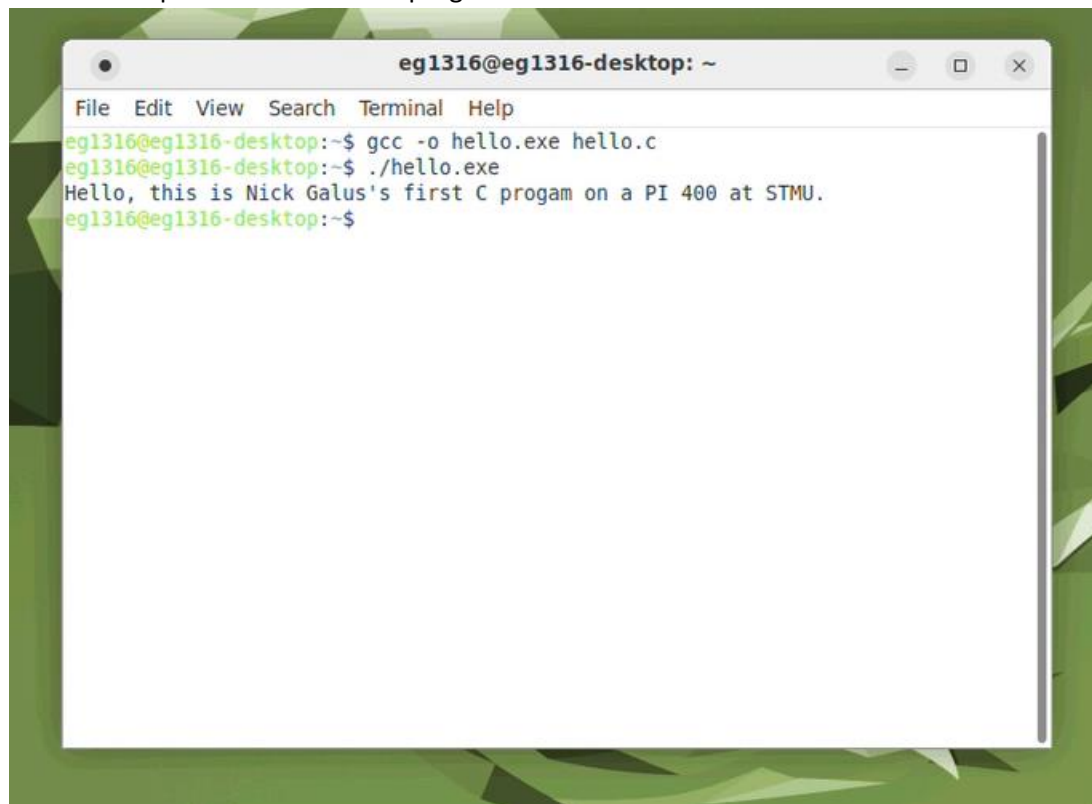
```
hello.c
File Edit View

#include <stdio.h>
int main()
{
    printf("Hello, this is Nick Galus's first C program on a PI 400 at STMU. \n");
    return 0;
}
```

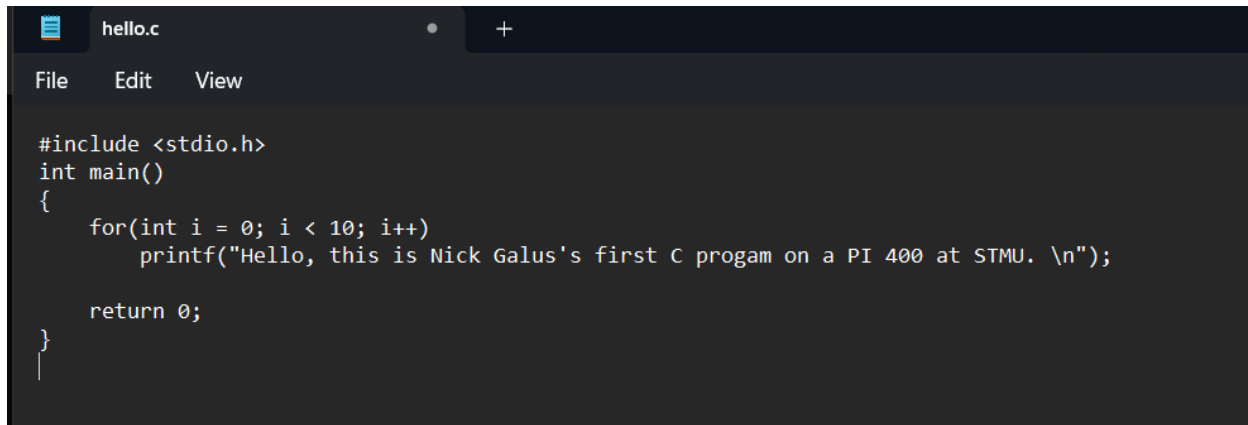
2. Transfer the “hello program” to the PI



3. Compile and execute the program within the terminal on the PI



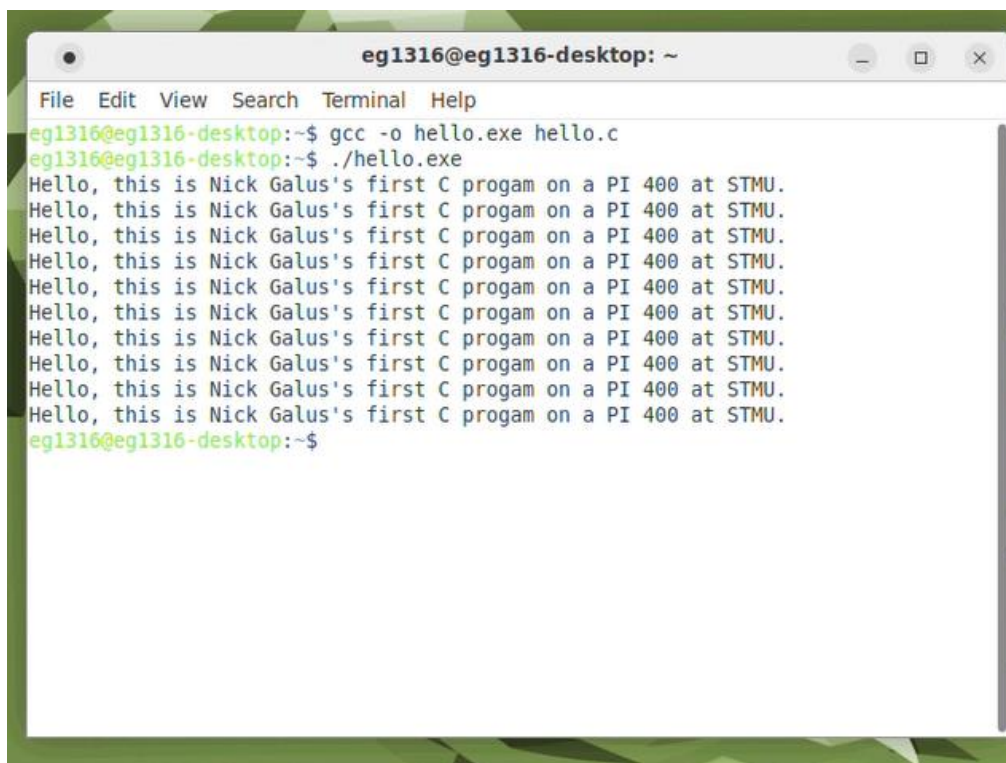
4. Modify the hello world code to print your name on the screen 10 times, then compile and execute on the PI



```
hello.c
File Edit View

#include <stdio.h>
int main()
{
    for(int i = 0; i < 10; i++)
        printf("Hello, this is Nick Galus's first C program on a PI 400 at STMU. \n");

    return 0;
}
```



```
eg1316@eg1316-desktop: ~
File Edit View Search Terminal Help
eg1316@eg1316-desktop:~$ gcc -o hello.exe hello.c
eg1316@eg1316-desktop:~$ ./hello.exe
Hello, this is Nick Galus's first C program on a PI 400 at STMU.
Hello, this is Nick Galus's first C program on a PI 400 at STMU.
Hello, this is Nick Galus's first C program on a PI 400 at STMU.
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Hello, this is Nick Galus's first C program on a PI 400 at STMU.
eg1316@eg1316-desktop:~$
```

7. Problems experienced and areas of concern

- Network file sharing was disabled by the security protocol
 - The resolution was to enable it in the security settings
- When running the C program from the Windows Command Prompt you get an error: 'gcc' is not recognized as an internal or external command, operable program or batch file
 - This occurs since GCC is not installed on the computer

